a Clinical comparisons

GSE286918 · 116 granulosa-cell samples
Stimulated IVF/ICSI · age × reserve

Primary: n=112 · mitochondrial fraction ≤20%
Excluded n=4; all 116 retained for sensitivity

Cross-cohort directional assessment

GSE193136 · DOR 6 / NOR 6

GSE232306 · DOR 6 / NOR 6

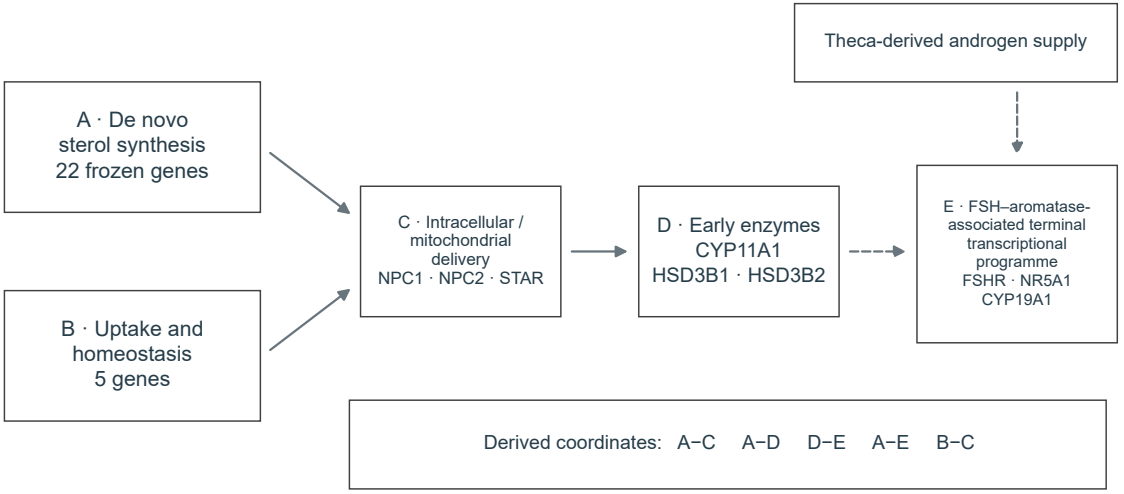
GSE274832 · DOR 3 / NOR 3

	Younger	Older
DOR	25	36
NOR	33	18

Younger <35 years; older ≥35 years

Separate protocol comparison
GSE191322 · poor responders
Conventional 5 / mild 5

b Five-node transcriptional framework



Solid: adjacent framework steps Dashed: indirect or external supply